

Teensy 4.1 Pin Usage on 4SQRP T41

4SQRP T41 Pin Usage		Teensy 4.1 Pin Reference												4SQRP T41 Pin Usage	
Ground	GND					GND	Teensy 4.1	VIN						VIN	5V Power
NC		PWM	CRX2	CS1	RX1	0		GND						GND	Ground
		PWM	CTX2	MISO1	TX1	1		3.3V							NC
Vol Encoder A/B	INT	PWM	OUT2			2		23	A9	CRX1	MCLK1	PWM	MCLK		Audio Adapter
	INT	PWM	LRCLK2			3		22	A8	CTX1		PWM	out		RXTX
FT Encoder A/B	INT	PWM	BCLK2			4		21	A7	RX5	BCLK1		BCLK		Audio Adapter
	INT	PWM	IN2			5		20	A6	TX5	LRCLK1		LRCLK		Audio Adapter
ADC Out	I2S RX (2)	PWM	OUT1D			6		19	A5		SCL	PWM	SCL		Audio Adapter
Audio Adapter	DIN	PWM	OUT1A		RX2	7		18	A4		SDA	PWM	SDA		Audio Adapter
Audio Adapter	DOUT	PWM	IN1		TX2	8		17	A3	TX4	SDA1		in		Tune Encoder B
NC	(4)	PWM	OUT1C			9		16	A2	RX4	SCL1		INT		Filter Encoder A/B
RA8875 CS	CS	PWM	MQSR	CS		10		15	A1	RX3	SPDIF IN	PWM	INT		Filter Encoder A/B
RA8875 MOSI	MOSI	PWM	CTX1	MOSI		11		14	A0	TX3	SPDIF OUT	PWM	in		Tune Encoder A
RA8875 MISO	MISO	PWM	MQSL	MISO		12		13	LED	SCK		PWM	SCK		RA8875 SCLK
NC						3.3V	Teensy 4.1	GND					GND		Ground
		PWM	SCL2	TX6	A10	24		41	A17						NC (1)
NC (1)		PWM	SDA2	RX6	A11	25		40	A16						
				MOSI1	A12	26		39	A15	MISO1	OUT1A		in		Busy SM Button
				SCK1	A13	27		38	A14	CS1	IN1		out		Mute
Filter 20m	out	PWM		RX7		28		37		CS		PWM	in		PTT
Filter 15m	out	PWM		TX7		29		36		CS		PWM	INT		Keyer Tip/Ring
Filter 80m	out		CRX3			30		35		TX8			INT		
Filter 40m	out		CTX3			31		34		RX8					
DAC DIN	I2S TX (3)		OUT1B			32		33			MCLK2	PWM			NC (1)

Color Key	PWM	Digital Audio	Serial	Digital	Analog	I2C	SPI	CAN Bus	Power
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NC = No Connection; INT = Interrupt; in/out = Digital input/output

- (1) Addition of pin header to Main board is possible
- (2) For default I2S pin usage see <https://www.pjrc.com/teensy/gui/?info=AudioInputI2SQuad>
Pin assignment is handled in AudioInputI2SQuad::begin
See pg 869 of IMXRT1060 manual (https://www.pjrc.com/teensy/IMXRT1060RM_rev3.pdf)
- (3) For default I2S pin usage see <https://www.pjrc.com/teensy/gui/?info=AudioOutputI2SQuad>
Pin 32 is AudioOutputI2SQuad TX ch 3+4 (I couldn't trace this assignment in the Teensy Audio library)
See pg 516 of IMXRT1060 manual (https://www.pjrc.com/teensy/IMXRT1060RM_rev3.pdf)
- (4) This pin is assigned to RA8875_RESET in the T41 code but is only used in the object construction.
The RA8875 library notes "In some modules you can leave unconnected but check if it's needed!"
It seems as that is what's been done on the T41.